

Chance Miller

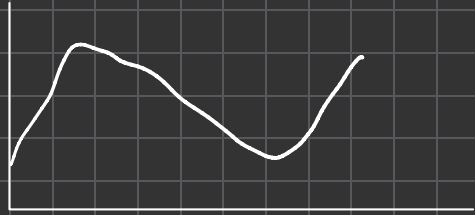
MICRO CONTROLLERS

Types of electrical signals:

- General form voltage through a wire relates to
- Voltage is usually compared to GND (0V)

Analogs:

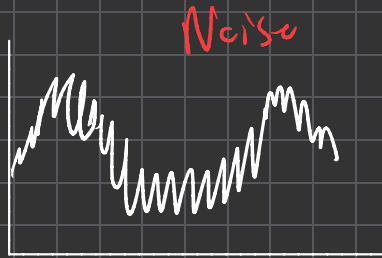
- Continuous Signal



- For our microcontrollers,

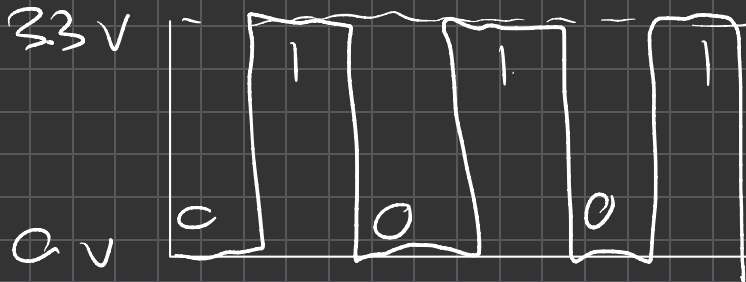
V ranges from 0-3.3V

- Problem: EASILY disrupted by noise



Digital Signals:

- Either $\sim$ 1 or 0, not continuous



- Low: GND, High: Component's max voltage
- Usually follows strict formations and timings to make sure components are speaking the same language

Reading Signals:

- Multimeter: Used to read voltage w/ various functions
- Oscilloscope: Visualizes voltage waveforms over time
- Microcontroller: Can read and write voltages to AC or DC signals

Form of a Microcontroller:

• What is a Microcontroller:

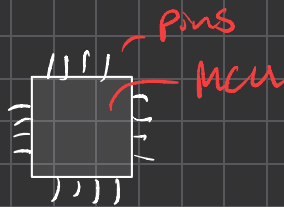
• Mini PC that sits on a PCB

• Connects via pins with signals and receivers

• Pins:

• Connects to traces on a PCB

• Each has a different name as shown in a Pinout



CAN BUS Basics:

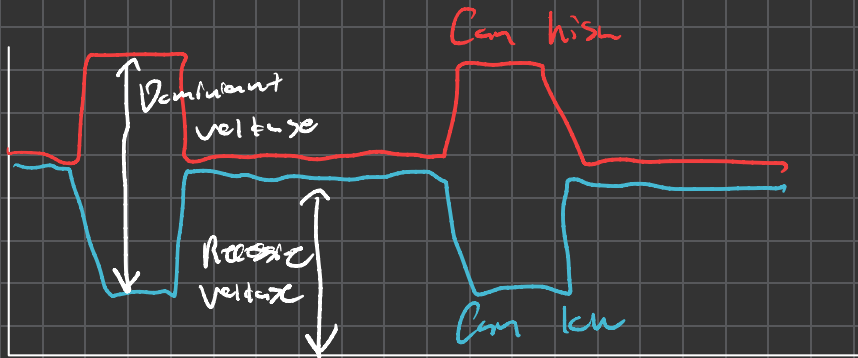
• CAN BUS Communication:

• Way to communicate using digital signals over environments with significant disturbance

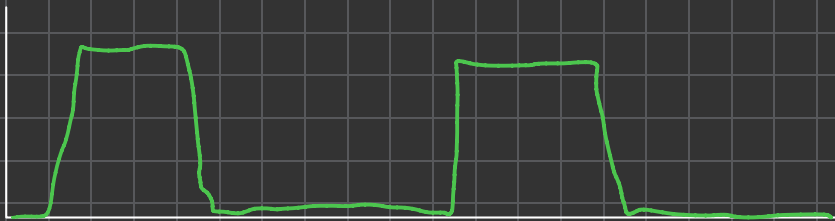
• Many MCUs can join the same CAN bus

Can Waveform!

- Emulates High and Low Signal level w/ 2 separate signals



CAN Message Signal



Driver Logic

- 0 is Sent $\rightarrow$ Both wires remain at same recessive voltage
- 1 is Sent $\rightarrow$ Both wires go to their dominant voltage
- When Voltages are subtracted, disturbances are heavily mitigated

CAN Vortex Packets:

- ID: Message ID unique to each byte of message
- VLL: Specifies length of data field
- DATA: All the binary data being sent in in the message, up to 8 bytes

Arbitration:

- CAN devices time synchronize to start transmissions in predefined windows
- CAN message ID's represent priorities
- A system called Arbitration allows for packet-prioritization w/o overhead